

Does Your Gut Really Affect Your Brain?

“... My gut feeling said so.” Almost all of us have heard this phrase at least once or twice in our lives, but have you ever wondered what our gut⁽¹⁾ has to do with actions that our brain controls entirely? Did you know that our gut is also famously known as the “second” brain?

What exactly are gut feelings, though? The gut often provides an outlet for emotions. The relationship of ‘soma to psyche’⁽²⁾ is perhaps greater than many will admit, even though it is apparent that our guts react to events in our environment. Almost everybody has experienced a queasy stomach⁽³⁾, loose bowel movements, or nausea due to various situations, whether it be an upcoming meeting, an exam you’re not ready for, or confessing your feelings to somebody. It is a common experience that contemplation of a difficult situation may elicit more gut reaction than the situation itself.

What exactly is the gut? In simple words, your gut is your gastrointestinal system and includes your stomach, intestines and colon. It digests and absorbs nutrients from food and excretes waste.

SOURCE: The Gut Stuff



You may wonder what exactly the gut has to do with our brain and why neuroscientists across the world study something

that, according to many people, is not even connected to the brain or how it functions!

It would be incorrect to say that the gut system is not connected with the brain; normally, the gut works silently, accepting whatever we feed it and seldom intruding into our consciousness except to signal hunger, satiety⁽⁴⁾, and the need to defecate. Its silent workings are governed by the enteric nervous system (ENS)⁽⁵⁾, with its complex network of nerves, hormones, and neurotransmitters⁽⁶⁾.

Then you might wonder, why is it called the “second brain”? ENS, or "gut brain," is intricately connected with the central nervous system or brain. Thus, the mechanisms are in place for the brain to sense events in the gut and to alter gut function. ENS uses the same chemicals and cells as the brain to help us digest and to alert the brain when something is amiss. The ENS is two thin layers of more than 100 million nerve cells lining your gastrointestinal tract from the oesophagus to the rectum.

How does this whole thing even work out? Our bodies are among the most complex systems ever made! Did you know that there is an entire pathway through which the ENS interacts with the central nervous system (CNS)? It is known as the gut-liver-brain axis. The gut-liver-brain axis is a three-way highway of communication. The connection between the gut and the liver lies in the gut barrier, whose disruption leads to more bacteria or their metabolites entering

the liver. Research suggests that digestive-system activity may affect cognition (thinking skills and memory), too.

Were you aware that the human gut microbiome⁽⁷⁾ impact human brain health in numerous ways? Gut microbes can produce hormones and neurotransmitters that are identical to those produced by humans. Bacterial receptors for these hormones influence microbial growth and virulence. Gut bacteria directly stimulate afferent⁽⁸⁾ neurons of the enteric nervous system to send signals to the brain via the vagus nerve⁽⁹⁾. Through these varied mechanisms, gut microbes shape the architecture of sleep and stress reactivity of the hypothalamic-pituitary-adrenal axis⁽¹⁰⁾. They influence memory, mood, and cognition and are clinically and therapeutically relevant to a range of disorders, including alcoholism, chronic fatigue syndrome, fibromyalgia⁽¹¹⁾, and restless legs syndrome⁽¹²⁾.

If one catches a disease which affects this axis, it not only impacts one's physical health but could cause harm to one's mental peace and well-being, such as an increase in stress, panic attacks, anxiety, chronic depression, increased chances of strokes and irritable bowel movements, along with obesity.

Therefore, eating healthy scientifically impacts not only your physical health by keeping your gut healthy but also plays a key role in the development of your mental health and well-being. In a nutshell, the human gut system is far more complex than it may seem, and the microbiome has a far greater effect on us than we realise. So, the next time you say “..my gut feelings,” remember that it is just your second brain sending signals to you (indirectly), like your own guardian angel trying to help you distinguish the right from the wrong!

KEY TO MEANINGS :

1. Gut	<u>The tube in your body that food passes through when it leaves your stomach</u>	7. Microbiome	<u>The collection of all microbes, such as bacteria, fungi, viruses, and their genes that naturally live on our bodies and inside us</u>
2. Soma and Psyche	<u>Soma: The Greek word for the physical body</u> <u>Psyche: The Greek word for the mind, soul, or spirit.</u>	8. Afferent	<u>Afferent neurons, also called sensory neurons, are nerve fibres that carry sensory information from the outside world into the brain.</u>

3. Queasy Stomach	<u>Nausea; feeling an urge to vomit</u>	9. Vagus nerve	<u>Also known as the vagal nerves , they are the main nerves of your parasympathetic nervous system This system controls specific body functions such as your digestion, heart rate and immune system</u>
4. Satiety	<u>the feeling of fullness and the suppression of hunger after eating</u>	10. Hypothalamic- pituitary-adrenal axis	<u>a communication system between three organs. It's crucial for your body's stress management. These endocrine system organs create a feedback loop of hormones to enact and regulate your body's stress reaction.</u>

5. ENS	<u>The enteric nervous system (ENS) is a neural network within the gastrointestinal (GI) tract that functions independently and controls various aspects of GI physiology. It consists of three major plexuses and communicates bidirectionally with the central nervous system (CNS) through intrinsic and extrinsic innervations.</u>	11. fibromyalgia	<u>Fibromyalgia is a chronic pain disorder primarily characterized by widespread musculoskeletal pain, often accompanied by fatigue, cognitive dysfunction, sleep disturbances, and psychiatric symptoms such as anxiety and depression.</u>
6. Neuro-Transmitters	<u>Endogenous chemicals that allow neurons to communicate with each other</u>	12. Restless leg Syndrome	<u>Restless legs syndrome is a common condition where you have an urge to move your legs to</u>

	<u>throughout the body.</u>		<u>relieve unpleasant sensations.</u> <u>Symptoms of restless legs syndrome are usually worse when you're trying to fall asleep and can prevent you from getting enough sleep.</u>
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SOURCES:

1. Johns Hopkins Medicine
2. UCLA Medical School
3. Healthline
4. Nature Journal
5. National Institute of Health (NIH)
6. Science Direct
7. Harvard Medical School

Author's

Note:

Dear reader, I hope that you enjoyed reading this paper curated with love. As somebody with not a very good diet (and to all those who may be stuck in the same situation as I was), learning and educating myself on such a topic definitely has changed my views about eating healthy, listening to my body, and taking care of my mental well-being as well. I had always wondered what the gut even had to do with our decision-making

and critical thinking skills; I believed it was just one of those idioms people used unknowingly until I learned about such amazing facts about our gut, brain, and how they interact with each other. I hope that this study succeeded in introducing you to such a complex topic. Thank you for reading, and I hope that you had a great read!

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